SPECTRA-SHIELD V1

Improved Proposal - Selective RF Interference Evaluation and a Protected, Authenticated UAV Telemetry Link

Team: 4 engineers
Duration: 6-8 weeks
Budget: INR 45,000-80,000 target; this plan uses ~INR 29,100
Setting: Indoor RF lab, conducted (cabled) prototype only
Interference tier: Tier 1 only
Friendly link: 865-868 MHz LoRa (SX1262 / SX1276, CSS)
Deliverables: Improved proposal document + code scaffolding

1. Locked design

- Friendly link: LoRa SX1262 on ESP32-S3 (Heltec WiFi LoRa 32 V4), CSS, 865-868 MHz India license-exempt (25 mW e.r.p., 1% duty).
- Interferer (Tier 1): CC1101 868 MHz module on an ESP32, emitting OOK/FSK CW plus same-channel noise on the LoRa channel.
- Harness: all-cabled / conducted - friendly TX -> fixed att -> 2-way combiner -> friendly RX, and interferer -> step att -> same combiner.
- Monitoring: RTL-SDR Blog V4 to independently confirm interferer frequency/level and containment (no open-air radiation).
- Tier 2 (HackRF One) and Tier 3 (ADALM-PLUTO) dropped from the default plan as too costly; kept as optional upgrade notes.

2. Bill of materials (verified India prices, 2026)

Item	Qty	Unit INR	Total INR
Heltec WiFi LoRa 32 V4 (ESP32-S3 + SX1262)	2	3549	7098
CC1101 868 MHz module (interferer)	2	250	500
SMA step attenuator 0-10 dB (1 dB step)	1	2500	2500
Fixed SMA attenuators (6/10/20 dB set)	~4	1500	6000
2-way power combiner / divider	1	2000	2000
SMA cables / adapters / pigtails	-	-	1500
RTL-SDR Blog V4 (monitor / verify)	1	6028	6028
Breadboard / PSU / enclosure / mounts	-	-	2000
Consumables / misc	-	-	1500
TOTAL			**29100**

Budget remaining vs the INR 45,000 floor: ~INR 16,000 (spare radios, second interferer, or report printing). Vendors referenced: zbotic.in, Robu.in, compoindia, Synergy Telecom / Dipeshwari.

3. Test methodology (conducted)

- Calibrate the harness: measure path loss per attenuator setting using the RTL-SDR and SX1262 RSSI.
- Sweep J/S via the step attenuator in 1 dB steps; per point record PER, RSSI, SNR, p95 latency, jitter, and auth accept/reject.
- Headline metric: J/S at 10% PER, plus a full PER-vs-J/S curve.
- Demonstrate LoRa resilience against an un-coded baseline (optional: CC1101 link as the fragile comparison).

4. Firmware + authentication design

- Frame: hdr | dev_id | seq | timestamp | payload | CRC16 | HMAC-SHA256 (truncated 16 B). Verify CRC before HMAC.
- Replay defense: monotonic counter + timestamp window; optional ground-nonce challenge-response.
- Crypto: ESP32-S3 hardware AES (~7.3 MB/s) and SHA-256 (~26 MB/s); mbedTLS with CONFIG_MBEDTLS_HARDWARE_AES/SHA, or the ESP-IDF HMAC peripheral with an eFuse-burned key (one-shot, fine for small frames).
- Payload: CBOR/MsgPack to minimize airtime (duty-cycle compliance).
- Provisioning: per-device keys via build-time secrets, never hardcoded in source.

5. Software

- Logging: SQLite + CSV.
- Dashboard: Streamlit (PER/J/S plot, RSSI/SNR, latency histogram, auth counters).
- Analysis: Python / Matplotlib for report figures.

6. Repository layout

```
docs/       01-proposal 02-architecture 03-test-methodology 04-bom 05-safety-legal
firmware/   common/ (frame, auth, cbor)  uav_node/  ground_node/  interferer/
host/       logger/ (sqlite, csv)  analysis/ (PER plots)  dashboard/ (streamlit)
hardware/   conducted_setup  calibration
README.md   platformio.ini   requirements.txt
```

Firmware target: PlatformIO + Arduino core + RadioLib (SX1262). ESP-IDF noted as an option for the hardware-HMAC path.

7. Schedule (6-8 weeks, 4 engineers)

Week	Work
W1	Order BOM; bench + harness calibration
W2	Friendly link bring-up; frame format
W3	Auth: HMAC, counter/replay; provisioning + replay-attack demo
W4	Interferer (CC1101 CW / same-channel); full conducted chain
W5	J/S sweep runs; data capture
W6	Dashboard + analysis + figures
W7	Repeatability, edge cases, report draft
W8	Buffer, demo, presentation

Team split: 2x firmware, 1x RF/test, 1x software/analysis.

8. Safety & legal

- Conducted-only; verify no radiation with the RTL-SDR outside the rig.
- Operate only in 865-868 MHz and 2.4 GHz ISM license-exempt bands. Never GPS or cellular bands.
- Low EIRP behind attenuators; interference research limited to a closed, cabled setup.
- India WPC rule: 'Use of Low Power Equipment in the Frequency Band 865-868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021', G.S.R. 853(E), 10 Dec 2021 (865-868 MHz, 25 mW e.r.p., 1% duty cycle).

9. Open risks

- CC1101 sometimes out of stock -> fallback: a second LoRa module as interferer.
- Fixed attenuators have MOQ 5-10 -> buy a small set from one vendor, or lean on the step attenuator + one 20 dB pad.
- ESP32-S3 hardware HMAC is one-shot (no streaming) -> fine for our small frames.